

“The whole future lies in
uncertainty: live immediately.”

- Seneca

StoicQuotes.com



class starts @ 9:04pm

Agenda

- ① kafka consumer group
- ② kafka demo
- ③ Docker

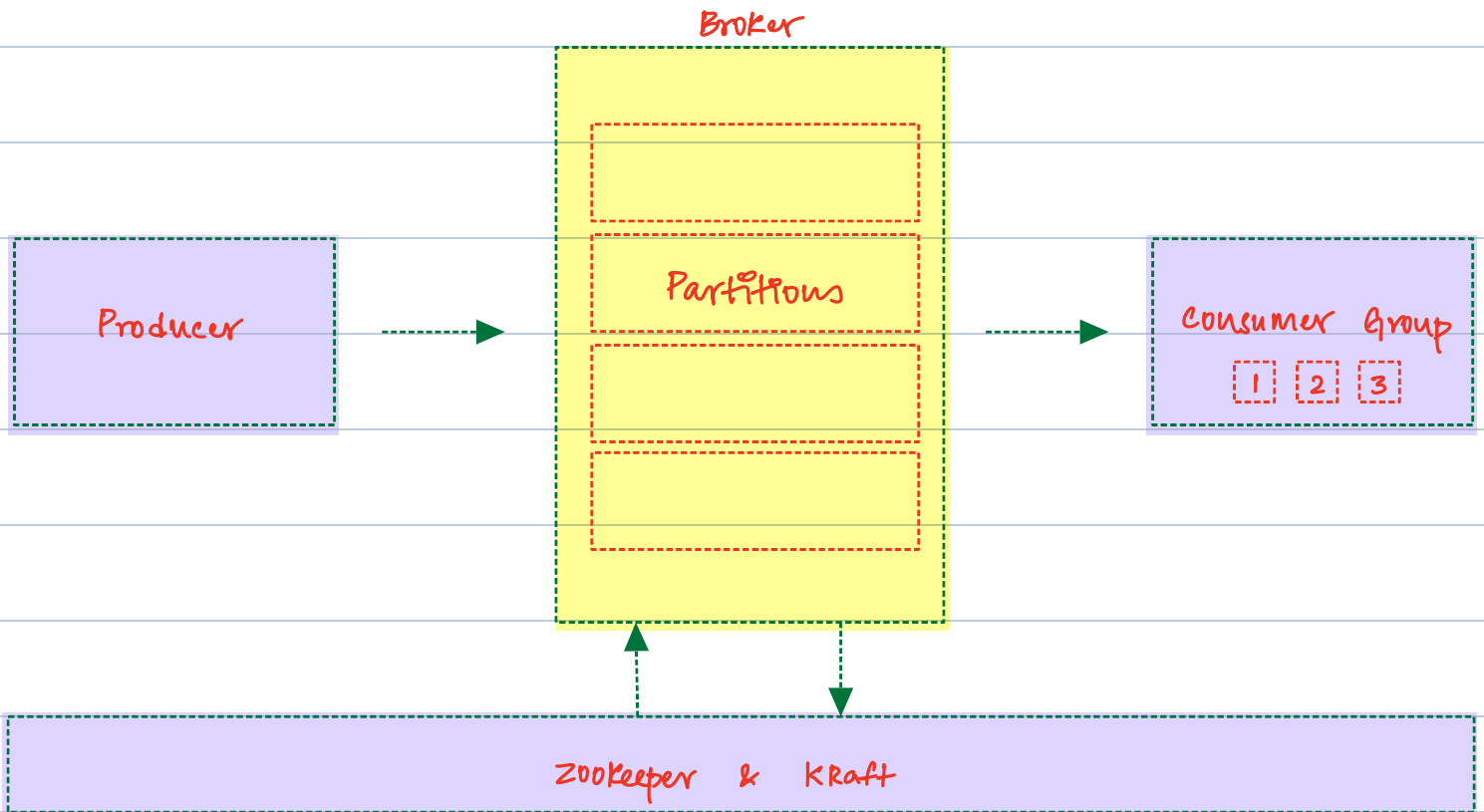
KAFKA

- ① Introduction to Airflow
- ② Airflow Components
- ③ Demo Airflow

AIRFLOW

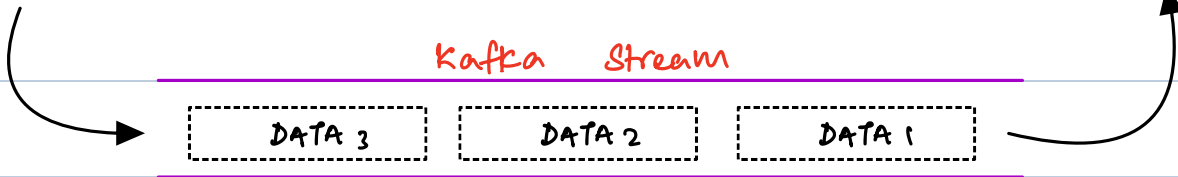
Pager Duty → Any incident, give me 2 minutes.

Kafka Consumer Group & Architecture



JPo Hotstar

India vs Aus



JPo Hotstar :

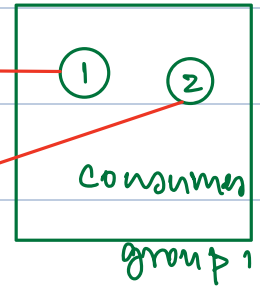
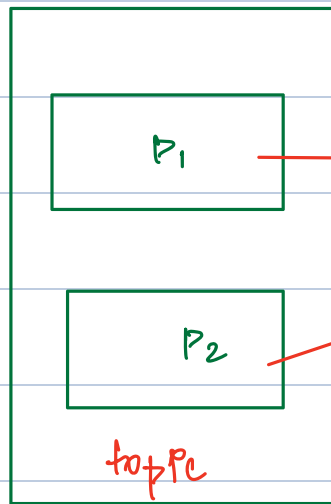
User mobile :

① How is partitioning helping with throughput ?

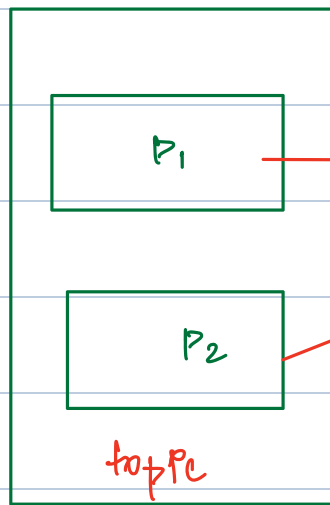
② What is the default behaviour of consumer groups & Partitions ?

③ Kafka sequential write.

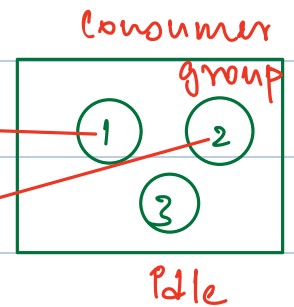
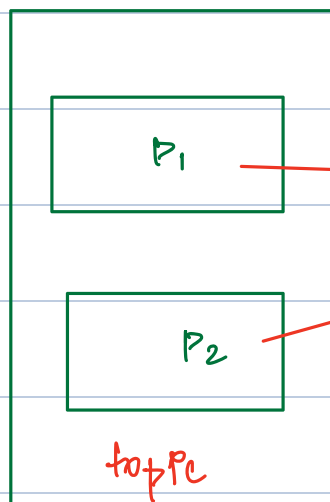
stock
market



stock
market



stock
market



Kafka Demo



Binance API: Reading real time stream data

Kafka: Async communication

Spark streaming: Transforming stream data

Influx DB: Storing time series data

Grafana: Creating line chart

Introduction to Airflow

Extract Transform Load

Usecase:

- ① Sync data from customer Database everyday
- ② Enhance, dedupe and store it
- ③ Generate custom dashboards and alerts

Traditional solution

- ① Cron Job
- ② Manually Retrigger

Questions to answer

- ① What should we do if step 2 fails?
- ② How to set retry configuration?
- ③ How to notify if something failed or succeeded
- ④ How to monitor them and store logs?
- ⑤ How to handle complex dependencies?

What is Airflow?

Open source platform for developing, scheduling and monitoring

batch oriented workflows

- ① Designed for orchestrating complex workflows
- ② Built using python, you can define workflows or DAGs as code
- ③ Open source - primarily developed by Airbnb

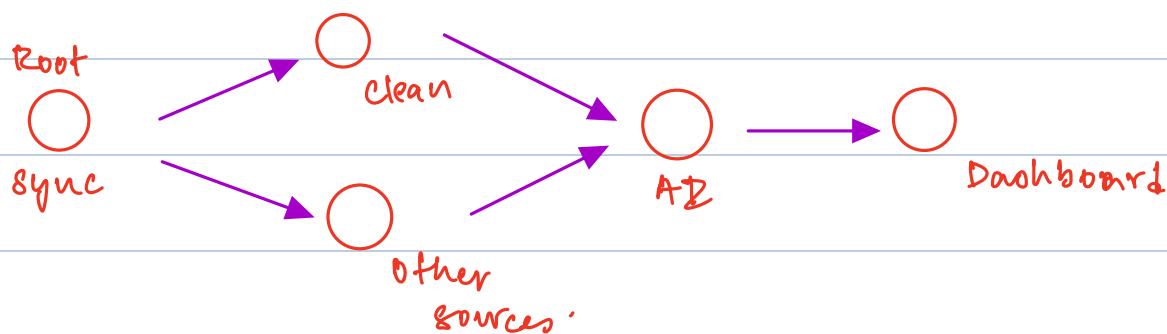
Airbnb + workflow = Airflow

Features

- ① Orchestrate Batch Jobs
- ② Scalable
- ③ Extensible
- ④ UI for monitoring

DAGs - Heart of your Airflow

Directed Acyclic Graph → code, magic, glue



children of clean: AI

parents of clean: sync

Airflow Components

Dag encapsulates

① schedule

③ Task Deps

② Task

smallest unit
of work

④ callback

→ notification when
DAG is successful

Components

① scheduler

② Task

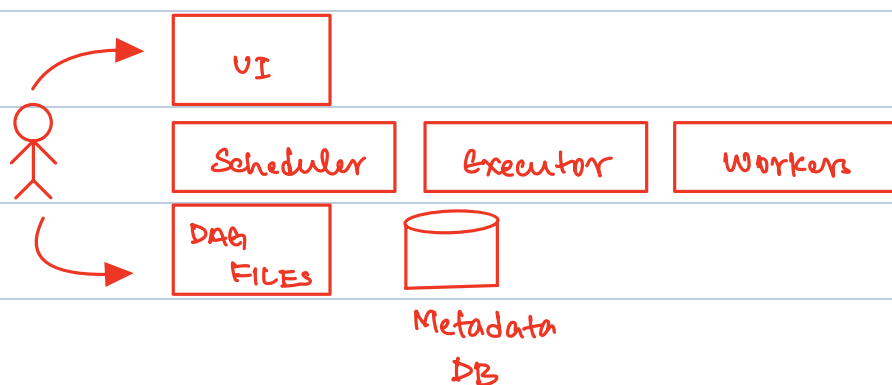
③ Executor [k8, celery]

④ DAG File Processor

⑤ UI component

⑥ Metadata DB [SQLite, postgres, MySQL]

⑦ Workers



Demo Airflow